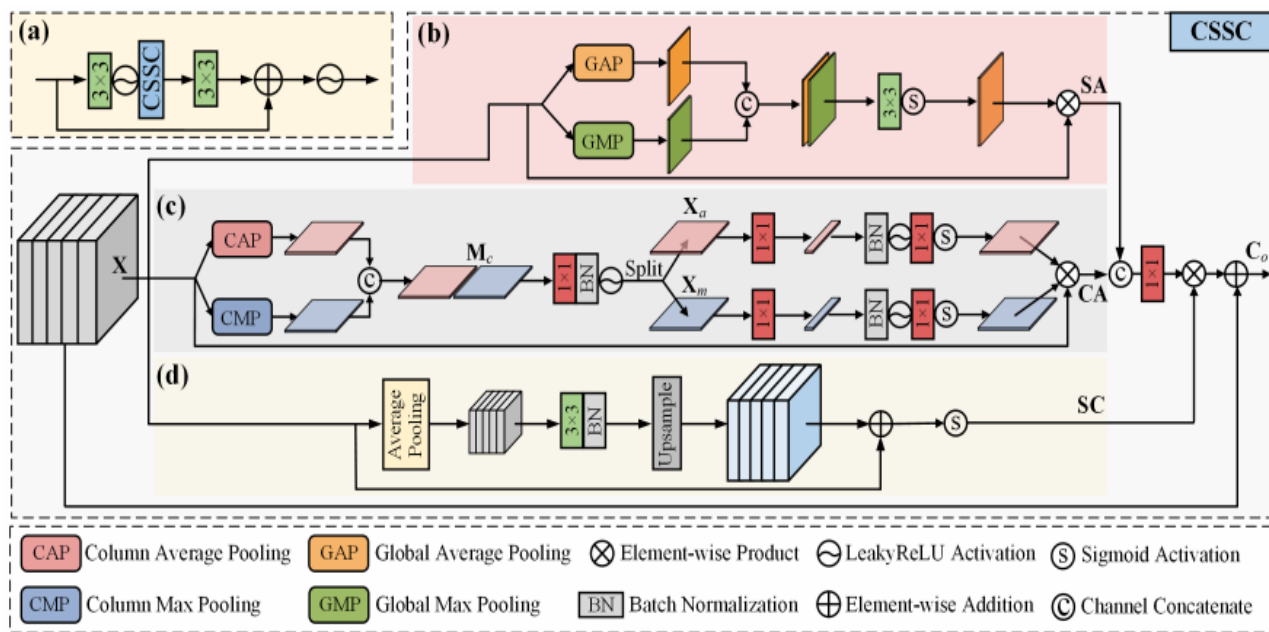


论文信息



论文题目: ASCNet: Asymmetric Sampling Correction Network for Infrared Image Destriping

中文题目: ASCNet: 用于红外图像条带校正的非对称采样校正网络

论文链接: <https://ieeexplore.ieee.org/document/10855453>

官方 github: <https://github.com/xdFai/ASCNet/tree/main>

所属机构: 西安电子科技大学光电工程学院等

核心速览: 本文介绍了一种名为非对称采样校正网络 (ASCNet) 的红外图像条纹去除方法, 该方

法能够有效捕获全局列关系并嵌入到 U 型框架中，提供全面的判别性表示和无缝的语义连接。

RCSSC: 在长距离依赖中捕获列关系，通过结合列、空间和自依赖信息，建立全局上下文以区分条纹和场景的垂直结构。

```
RCSSC(
  (head): Sequential(
    (0): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (1): LeakyReLU(negative_slope=0.01)
  )
  (SC): Sequential(
    (0): AvgPool2d(kernel_size=4, stride=4, padding=0)
    (1): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (2): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  )
  (SA): spatial_attn_layer(
    (compress): ChannelPool()
    (spatial): Basic(
      (conv): Conv2d(2, 1, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
    )
  )
  (CA): CALayer(
    (avgPoolW): AdaptiveAvgPool2d(output_size=(1, None))
    (maxPoolW): AdaptiveMaxPool2d(output_size=(1, None))
    (conv_1x1): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (bn): BatchNorm2d(64, eps=1e-05, momentum=0.01, affine=True, track_running_stats=True)
    (Relu): LeakyReLU(negative_slope=0.01)
    (F_h): Sequential(
      (0): Conv2d(32, 2, kernel_size=(1, 1), stride=(1, 1))
      (1): BatchNorm2d(2, eps=1e-05, momentum=0.01, affine=True, track_running_stats=True)
      (2): ReLU(inplace=True)
      (3): Conv2d(2, 32, kernel_size=(1, 1), stride=(1, 1))
    )
    (F_w): Sequential(
      (0): Conv2d(32, 2, kernel_size=(1, 1), stride=(1, 1))
      (1): BatchNorm2d(2, eps=1e-05, momentum=0.01, affine=True, track_running_stats=True)
      (2): ReLU(inplace=True)
      (3): Conv2d(2, 32, kernel_size=(1, 1), stride=(1, 1))
    )
    (sigmoid): Sigmoid()
  )
  (conv1x1): Sequential(
    (0): Conv2d(64, 32, kernel_size=(1, 1), stride=(1, 1))
    (1): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  )
  (ReLU): LeakyReLU(negative_slope=0.01)
  (tail): Conv2d(32, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
)
微信公众号: AI缝合术!
输入张量的形状: torch.Size([1, 32, 256, 256])
输出张量的形状: torch.Size([1, 32, 256, 256])
```